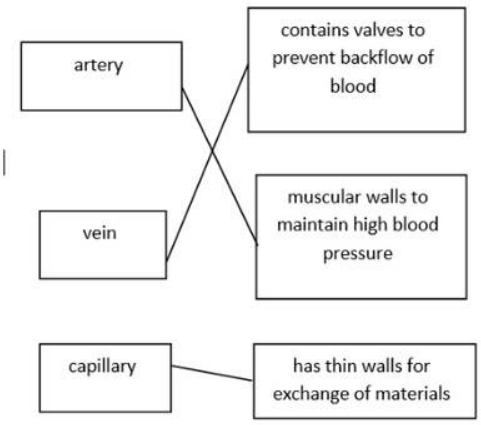
Type MS · Identify blood components and match them to their features and jobs

Answer 1 of 14 · Question 2 part a.i · 3 marks

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

2 (a) (i)	X = plasma; Y = red blood cell; Z = white blood cell/phagocyte;	Accept named phagocyte	(3)
(a) (ii)	• transports oxygen; • to body/ cells/tissues/organs;		(2)
(a) (iii)	the patient has an infection/leukaemia/allergy/ blood cancer, inflammation;	Accept pathogens in body	(1)
(b)		Reject more than one line from each blood vessel 3 correct=2marks 1 or 2 correct= 1mark	(2)
(c) (i)	left ventricle;		(1)
(c) (ii)	• into right ventricle; • through pulmonary artery; • to the lungs; • (from the lungs) into the pulmonary vein;		(4)

Total for Question 2 = 13 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 2 of 14 · Question 1 part b · 2 marks

2020 NOV · Paper 01 (regional R) · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Identify blood components and match them to their features and jobs

Answer 3 of 14 · Question 3 part a.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a) (i)	C; (red blood cells)		1
(ii)	any three from • more carbon dioxide (1) • less oxygen (1) • moister (1) • warmer (1)		3
(iii)	• large surface area (1) • thin (walled)/wall one cell thick (1) • good blood supply (1) • moist lining (1)		3
(b)	• thin wall (1) • narrow lumen (1)	Allow references to only one blood cell passing through	2
(c)	• breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1) • as lungs would not squeeze out air/forced exhalation(1)		2

Total 11 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 3 of 14 · Question 3 part a.i · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 4 of 14 · Question 6 part b.ii · 4 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Question number					
6	(a)	(i)	phagocytosis;		1
		(ii)	• enclosed bacterium fuses with lysosome; • enzymes break down /digests bacterium; • remains of bacterium removed from white blood cell;		3
		(iii)	• label/identify/complementary to bacterium/antigens presented; • so response quicker if new infection/better immune response;		2
	(b)	(i)	any three from • prick skin to obtain blood; • use a sterile needle/pin • smear/spread blood on slide; • ensure smear is thin; • add stain; • allow to dry;		3
		(ii)	any four from • red blood cells would be seen as circle/not in 3D; • white blood cells would be larger than red/rbc too large; • wouldn't see platelets; • because too small for light microscope;		4
		(iii)	• no risk of infection; • e.g. AIDS;	R contaminate	2

Total 15

Type MS · Identify blood components and match them to their features and jobs

Answer 4 of 14 · Question 6 part b.ii · 4 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 5 of 14 · Question 7 part a · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

7 (a)	• plasma contains no cells (1) • plasma colourless/whole blood red (1)	Allow reverse for whole blood	2
(b)	any six from • blood of patient contains antibodies (1) • (antibodies) bind to Ebola/virus antigens/antibody-antigen complex formed/antibodies complementary to (virus) antigens (1) • phagocytes engulf (antibody-antigen complex) (1) • Ebola/virus destroyed (1) • testing of patient's blood ensures no disease present/prevents transfer of disease (1) • removal of red blood cells ensures no reaction when blood transfused/no agglutination (1) • anti-Ebola antibodies in transfused blood help person with disease to recover (1) • more effective than medicine (1) • quicker effect (1)	Ignore fight disease/virus Allow passive artificial immunity	6

Total 8 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 5 of 14 · Question 7 part a · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 6 of 14 · Question 2 part a.iv · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.iv in this mark scheme extract

Number				
2	(a) (i)	• genetic material made of RNA (1) • no DNA (1)		2
	(ii)	<table><tr><th>Disease</th><th>Blood Tested (✓)</th></tr><tr><td>anaemia</td><td></td></tr><tr><td>cystic fibrosis</td><td></td></tr><tr><td>gonorrhoea</td><td>✓</td></tr><tr><td>HIV</td><td>✓</td></tr><tr><td>scurvy</td><td></td></tr></table>	-1 for each extra tick	2
	(iii)	• donor/blood transfused into person with Ebola (1) • needs to be compatible/matched/same group/not rejected (1) • otherwise agglutination/clumping occurs (1)	R clotting	3
	(iv)	• cause blood cells to burst/cells are damaged/destroyed/lose structure (1) • results in loss of function (1)		2
	(v)	• antibodies in donated blood (1) • can destroy virus in infected person (1)		2
	(b)	• uracil present, not thymine (1) • guanine would pair with cytosine/adenine with uracil (1) • percentage of G and C/A and U would be the same (1)		3

Total for question = 14 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 6 of 14 · Question 2 part a.iv · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 7 of 14 · Question 1 part whole · 6 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

1

	Feature					
Blood component	**has a nucleus**	**contains haemoglobin**	**carries hormones**	**engulfs pathogens**	**produces antibodies**	**helps blood clotting**
red blood cell		✓				
phagocyte	✓			✓		
lymphocyte	✓				✓	
platelets						✓
plasma	;	;	✓ ;	;	;	;

(6)

Total 6 Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 7 of 14 · Question 1 part whole · 6 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Identify blood components and match them to their features and jobs

Answer 8 of 14 · Question 1 part whole · 4 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

Question number			
1	<table><tr><th></th><th colspan="4">Component</th></tr><tr><th>Feature</th><th>Red blood cell</th><th>Plasma</th><th>Platelet</th><th>Phagocyte</th></tr><tr><td>transports oxygen</td><td>✓;</td><td></td><td></td><td></td></tr><tr><td>has a nucleus</td><td></td><td></td><td></td><td>✓;</td></tr><tr><td>consists of 90% water</td><td></td><td>✓;</td><td></td><td></td></tr><tr><td>involved in blood clotting</td><td></td><td>✓</td><td>✓;</td><td></td></tr></table>	Mark by row	4

Total 4 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 8 of 14 · Question 1 part whole · 4 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 9 of 14 · Question 3 part a · 4 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

number				
3	(a)	<table><tr><th>Component</th><th>Name</th><th>Function</th></tr><tr><td>A</td><td>red blood cell/erythrocyte</td><td>carry/transport oxygen</td></tr><tr><td>B</td><td>white blood cell/phagocyte;</td><td>engulf bacteria;</td></tr><tr><td>C</td><td>platelets;</td><td>clotting;</td></tr></table>		4 marks
	(b) (i)	any five of • enzyme/thromboplastin/thrombokinase; • released from (damaged) platelets; • prothombin to thrombin; • fibrinogen to fibrin; • meshwork formed; • trapped rbc form clot; • ref. to Ca^{2+};	reject secrete	5 marks
	(ii)	• prevents loss of too much blood; • prevents entry of pathogens/bacteria; • which would cause infection/prevents infection;		3 marks

Total 12 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 9 of 14 · Question 3 part a · 4 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 10 of 14 · Question 1 part a · 5 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

1 (a)	<table><tr><th>Feature</th><th>red b c</th><th>phagocyte</th><th>lymphocyte</th><th>platelet</th><th>plasma</th></tr><tr><td>transports oxygen</td><td>✓;</td><td></td><td></td><td></td><td></td></tr><tr><td>contains haemoglobin</td><td>✓;</td><td></td><td></td><td></td><td></td></tr><tr><td>is a fluid</td><td></td><td></td><td></td><td></td><td>✓;</td></tr><tr><td>produces antibodies</td><td></td><td></td><td>✓;</td><td></td><td></td></tr><tr><td>engulfs pathogen</td><td></td><td>✓;</td><td></td><td></td><td></td></tr></table>	if more than required ticks no marks	5
(b)	any four from • forms a plug/mesh; • to prevent loss of (excess) blood; • prevents entry of bacteria/pathogens/microorganisms; • prevents infection/disease; • protects skin whilst healing;	R scab	4

Total 9 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 10 of 14 · Question 1 part a · 5 marks

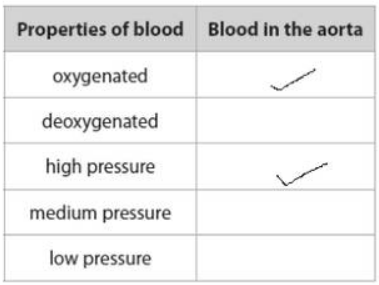
2023 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 11 of 14 · Question 2 part c.i · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

Number				
2	(a)	(i)	left ventricle;	1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves 2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick. 2
		(iv)	the lungs/pulmonary vein;	1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection 2
	(c)	(i)	C platelets;	1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;	2

Total 11 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 11 of 14 · Question 2 part c.i · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify blood components and match them to their features and jobs

Answer 12 of 14 · Question 5 part c · 1 mark

2024 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

5 (a)	(cardiac) muscle(s);		(1)
(b)	• pumps blood twice in one circuit; • right side pumps deoxygenated blood to lungs; • left side pumps oxygenated blood to body;		(3)
(c)	**C (lymphocytes and phagocytes)** The following are incorrect because: A platelets are not cells B antibodies and hormones are not cells D platelets are not cells		(1)
(d) (i)	hormones;		(1)
(ii)	platelets/fibrinogen/prothrombin;		(1)
(e)	(amount of blood in 1kg) 5.5 ÷ 70 = 0.07857; (amount of blood in 95kg) 0.07857 x 95 = 7.464; (number of red blood cells in 95kg) = 7.464 x 5.9 x 10^{12} = 4.4 x 10^{13};	allow 0.08 allow 7.5 allow 4.4038 ×10^{13} allow 4.425 × 10^{13} allow 4.484 × 10^{13} full marks for correct final answer	(3)

Total for Question 5 = 10 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 13 of 14 · Question 5 part d.ii · 1 mark

2024 JUN · Paper 02 (regional R) · find part d.ii in this mark scheme extract

5 (a)	(cardiac) muscle(s);		(1)
(b)	• pumps blood twice in one circuit; • right side pumps deoxygenated blood to lungs; • left side pumps oxygenated blood to body;		(3)
(c)	**C (lymphocytes and phagocytes)** The following are incorrect because: A platelets are not cells B antibodies and hormones are not cells D platelets are not cells		(1)
(d) (i)	hormones;		(1)
(ii)	platelets/fibrinogen/prothrombin;		(1)
(e)	(amount of blood in 1kg) 5.5 ÷ 70 = 0.07857; (amount of blood in 95kg) 0.07857 x 95 = 7.464; (number of red blood cells in 95kg) = 7.464 x 5.9 x 10^{12} = 4.4 x 10^{13};	allow 0.08 allow 7.5 allow 4.4038 ×10^{13} allow 4.425 × 10^{13} allow 4.484 × 10^{13} full marks for correct final answer	(3)

Total for Question 5 = 10 marks

Type MS · Identify blood components and match them to their features and jobs

Answer 14 of 14 · Question 5 part d.i · 1 mark

2024 JUN · Paper 02 (regional R) · find part d.i in this mark scheme extract

5 (a)	(cardiac) muscle(s);		(1)
(b)	• pumps blood twice in one circuit; • right side pumps deoxygenated blood to lungs; • left side pumps oxygenated blood to body;		(3)
(c)	**C (lymphocytes and phagocytes)** The following are incorrect because: A platelets are not cells B antibodies and hormones are not cells D platelets are not cells		(1)
(d) (i)	hormones;		(1)
(ii)	platelets/fibrinogen/prothrombin;		(1)
(e)	(amount of blood in 1kg) 5.5 ÷ 70 = 0.07857; (amount of blood in 95kg) 0.07857 x 95 = 7.464; (number of red blood cells in 95kg) = 7.464 x 5.9 x 10^{12} = 4.4 x 10^{13};	allow 0.08 allow 7.5 allow 4.4038 ×10^{13} allow 4.425 × 10^{13} allow 4.484 × 10^{13} full marks for correct final answer	(3)

Total for Question 5 = 10 marks